Contents

A Introduction **2**

B Minkowski space and space-like surfaces **2**
Space-like immersions. 3
Entire surfaces. 3
The Gauss map in $\mathbb{M}^{2,1}$ 5
$\mathbb{H}$ is a model for the Hyperbolic plane. 6

C The Gauss-Codazzi equations and the curvatures of S **7**
C.1 First and second fundamental forms 7
C.2 Mean curvature and Gaussian curvature expr 7

D Riemann surfaces and harmonic maps **7**
D.1 Preliminaries on Riemann surfaces 8
Complexification 8
Complex surfaces 8
Conformal metrics 8
Conformal structures on 2-surfaces 9
D.2 Uniformization theorem 10
D.3 Harmonic maps between Riemann surfaces 10
D.4 Energy density and Jacobian 10
D.5 Hopf differential 10
D.6 Holomorphicity of the Hopf differential 10
D.7 Bochner identities 10
D.8 Harmonic maps into the Poincaré disk 10

E CMC surfaces and harmonic maps **10**
E.1 Gauss map harmonic if and only if mean curvature is constant . 10
E.2 Definition of w 10
E.3 Expression of the induced metric in terms of w and the Hopf differential 10
E.4 Derivation of the main PDE 10
E.5 Reconstruction of the surface using the Gauss-Codazzi equations 10

F Classification of hyperbolic CMC cuts **10**
F.1 The mean curvature of the CMC immersed in $\mathbb{M}^{2,1}$ cannot be -1 10
F.2 The idea behind the classification 10
F.3 The map $\mathcal{A}$ 10
F.3.1 Hyperbolic CMC cuts and conformal type of the disk . . 10
F.3.2 From a surface to a holomorphic quadratic differential . . 10
F.4 inverting the map $\mathcal{A}$ 10
F.4.1 Reconstruction from the PDE data 10
F.4.2 Use of Gauss and Codazzi equations 10
F.4.3 Fundamental theorem of surface theory 10
F.5 Statement of the classification theorem 10

G Uniqueness for the PDE **11**
G.1 A generalized maximum principle 11
A boundedness result 15
G.2 The uniqueness theorem 19

H existence for the nonlinear PDE **24**

A Introduction

Goal of the work, maybe a simple example, some mention to preliminary information and a soft mention to what's the problem and the philosophy.

B Minkowski space and space-like surfaces

The environment we will work within is the Minkowki 3-space, defined as $\mathbb{R}^3$ endowed with the pseudometric

$$ds^2 = (dx^1)^2 + (dx^2)^2 - (dx^3)^2.$$

In order to distinguish the use of this metric from the standard euclidean metric of $\mathbb{R}^3$ we will use the notation $\mathbb{M}^{2,1}$. We will also denote the coordinates either as (x^1, x^2, x^3) or (x, y, t), choosing the notation according to convenience. We will also identify the slice $\{(x, y, 0)\}$ of $\mathbb{M}^{2,1}$ with $\mathbb{R}^2$ and with $\mathbb{C}$.

The space $\mathbb{M}^{2,1}$ is an example of a more general object called a pseudo-Euclidean vector space, a term that refers to a finite-dimensional vector space V endowed with a nondegenerate (possibly not positive-definite) symmetric bilinear form g. This kind of space serves as the "infinitesimal model" for pseudo-Riemannian manifolds, smooth manifolds endowed with a smooth nondegenerate $(0, 2)$-tensor. If the signature of this tensor is exactly $(n-1, 1)$, the manifold is called a Lorentzian manifold.
The following definition, borrowed from physics, will be useful later.

Definition. Given (M, g) a Lorentzian manifold and $p \in M$, we say that a vector $v \in T_pM$ is

- space-like if $\langle v, v\rangle > 0$,
- light-like if $\langle v, v\rangle = 0$ and $v \neq 0$,
- time-like if $\langle v, v\rangle < 0$.

We also call a subspace $P \subseteq T_pM$ space-like if every nonzero vector in P is space-like.

We will always denote with S a differentiable manifold of dimension 2 (a surface) immersed in $\mathbb{M}^{2,1}$, $\iota : S \hookrightarrow \mathbb{M}^{2,1}$. As usual, the immersion induces a nondegenerate bilinear form on S, the first fundamental form or pull-back pseudometric, which we indicate as $\mathbf{I} = \iota^*(g_{\mathbb{M}^{2,1}})$ and write in coordinates as $\mathbf{I} = g_{ij}dx^i dx^j$ (for simplicity we sometimes forget the immersion ι).

Space-like immersions.
The objects we will focus on are not arbitrary immersed surfaces, but the so-called space-like surfaces of $\mathbb{M}^{2,1}$. This term refers to immersed surfaces in $\mathbb{M}^{2,1}$ whose induced metric is positive-definite, and hence Riemannian. Equivalently, these are the immersed surfaces S such that, for every $p \in S$, the tangent plane T_pS is space-like.

A nice property of this kind of surface is that they are locally the graph of a function of the form $x^3 = f(x^1, x^2)$.
We can show this with a simple argument:
First, we assumed that the restriction of the Lorentzian metric to each tangent space T_pS is positive-definite so that it's a space-like plane. This can be rewritten as

$$T_pS \subset \mathscr{C} := \{v \in \mathbb{M}^{2,1} : \langle v, v\rangle \geq 0\} = \{t^2 \leq x^2 + y^2\} \ .$$

The set $\mathscr{C}$ is the outside-region of the double cone in $\mathbb{M}^{2,1}$ whose axis is $\{x^1 = 0,\ x^2 = 0\}$. Clearly $\mathscr{C}$ does not contain any vertical vector and a fortiori also T_pM doesn't. Therefore we can apply the implicit function theorem and express S locally near p as the graph of a function $t = f(x, y)$, where $f : U \subset \mathbb{R}^2 \to \mathbb{R}$. Where this local representation is valid, we can parametrize the surface as $(x, y, f(x, y))$, and obtain the local frame

$$V_x = (1, 0, \partial_x f(x, y)), \qquad V_y = (0, 1, \partial_y f(x, y)),$$

for which the first fundamental form is represented by the matrix

$$g = \begin{pmatrix} \langle V_x, V_x\rangle & \langle V_x, V_y\rangle \\ \langle V_y, V_x\rangle & \langle V_y, V_y\rangle \end{pmatrix} = \begin{pmatrix} 1 - f_x^2 & -f_x f_y \\ -f_x f_y & 1 - f_y^2 \end{pmatrix}.$$

Since the parametrization is regular, the condition that g be positive-definite is equivalent to

$$0 < \det g = 1 - (f_x^2 + f_y^2) = 1 - |\nabla f|^2,$$

and therefore $|\nabla f|^2 < 1$.

Entire surfaces.
Given a surface S immersed in $\mathbb{M}^{2,1}$, we can define a projection $\Pi : S \to \mathbb{R}^2$ by forgetting the time coordinate, $\Pi(x^1, x^2, x^3) := (x^1, x^2)$ for any $(x^1, x^2, x^3) \in S$. We call the surface S is entire when the map Π is surjective. If the surface is globally a graph of a function φ, this condition is exactly stating that the function φ is an entire map from $\mathbb{R}^2$ to $\mathbb{R}$.

The situation for space-like surfaces is actually a lot nicer than what we presented up to now, as the representation as a graph is not only local but actually global as we prove in the proposition below.

{prop:toppropr spimm}

Proposition 1. *Let Ω be a domain in $\mathbb{R}^2$ (that is, an open connected set) and let $\iota : \Omega \hookrightarrow S \subset \mathbb{M}^{2,1}$ be a space-like immersion. Suppose that S is complete with respect to the induced Riemannian metric. Then S is an entire graph and Ω is simply connected.*

Before laying out the proof let's recall a definition and a proposition, whose proof can be found in **??**, from the theory of coverings. Let M and N two differentiable manifolds and $f : M \to N$ a smooth map.

Definition B.1. We say that the map f satisfies the path lifting property if for every $p \in N$ and for every path $\alpha : I \to M$ such that $\alpha(0) = f(p)$ there exists a path $\tilde{\alpha} : I \to N$ such that $\tilde{\alpha}(0) = p$ and $f \circ \tilde{\alpha} = \alpha$. We can assume without loss of generality that the paths involved are differentiable.

Proposition 2. *If f is a local homeomorphism and satisfies the path lifting property, then it's a covering map.*

Proof of Proposition 1. The idea of the proof is to show that $\Pi \circ \iota$ is a global diffeomrphism. Clearly this implies that Π restricted to S is a diffeomeorphism on $\mathbb{R}^2$ and so that S is an entire graph. From that we obtain also that Ω is simply connected since it's diffeomorphic to $\mathbb{R}^2$.

Let's observe that the target of $\Pi \circ \iota$ is just $\mathbb{R}^2$, which is simply connected, and the domain is Ω, which is connected since it's a domain. We already proved, by showing that S is locally a graph, that the map $\Pi \circ S$ is a local homeomorphism.
Having met these conditions, if we show that $\Pi \circ \iota$ satisfies the path lifting property, we conclude that it is a smooth covering map whose domain is simply connected, hence a diffeomorphism.

Let's fix a point $z \in \Omega$ and a path $\gamma : [0, 1] \to \mathbb{R}^2$ of bounded length, $\ell(\gamma) < \infty$, such that $\gamma(0) = (\Pi \circ \iota)(z)$. Our goal is to show that γ admits a lift to a path γ^* on Ω starting at z.
To this end, let's consider the set of times up to which such a lift exists,

$$\mathcal{J} := \{\tau \in [0, 1] \mid \exists \gamma^* : [0, t] \to \Omega \text{ such that } (\Pi \circ \iota) \circ \gamma^* = \gamma \text{ and } \gamma^*(0) = z\},$$

and let's define $\bar{\tau} := \sup \mathcal{J}$.

Let's observe that $0 \in \mathcal{J}$ and that $\mathcal{J}$ is open. This is true because, being $\Pi \circ \iota$ a local homeomorphism, if a certain τ is in $\mathcal{J} \cap (0, 1)$ then the whole interval $(\tau - \varepsilon, \tau + \varepsilon)$ for $\varepsilon > 0$ sufficiently small, belongs to it.

If we prove that $\bar{\tau} = 1$, then γ admits a lift on the whole interval $[0, 1]$ and we have finished. Then let's suppose by contradiction that $\bar{\tau} < 1$ and define $\gamma^* : [0, \bar{\tau}) \to \Omega$ the lift of γ up to $\bar{\tau}$. This curve exists by definition of $\bar{\tau}$.

Finally, we need to prove that the limit $\lim_{\tau \to \bar{\tau}} \gamma^*(t)$ exists. If this is true we can then extend γ^* up to $\bar{\tau}$, which means that $\bar{\tau} \in \mathcal{J}$. Since $\mathcal{J}$ is open, there exists an element $\bar{\tau} + \varepsilon \in \mathcal{J},\ \varepsilon > 0$. This contradicts the definition of $\bar{\tau}$ and proves $\bar{\tau} = 1$ since the alternative is absurd.

To prove that $\lim_{\tau \to \bar{\tau}} \gamma^*(\tau)$ exists we just need a couple of observations:

First we note that the map Π dilates distances, as

$$\begin{aligned}\langle v, v\rangle_{\mathbb{M}^{2,1}} &= (v^1)^2 + (v^2)^2 - (v^3)^2 \\ &\leq (v^1)^2 + (v^2)^2 \\ &= \langle \Pi v, \Pi v\rangle_{\mathbb{R}^2}\,.\end{aligned}$$

From this it's clear that $\ell(\gamma^*) \leq \ell(\gamma)$.
Now, if we fix any sequence $\tau_k \to \bar{\tau}$ and observe that as γ is differentiable, it's derivative is bounded, so $\ell(\gamma|_{[\tau_k,\tau_m]}) \leq C|\tau_k - \tau_m|$. By observing that

$$\begin{aligned}d(\gamma^*(\tau_k), \gamma^*(\tau_m)) &\leq \ell(\gamma^*|_{[\tau_k,\tau_m]}) \\ &\leq \ell(\gamma|_{[\tau_k,\tau_m]}) \\ &\leq C|\tau_k - \tau_m|\end{aligned}$$

we deduce that also $(\gamma^*(\tau_k))_k$ is a Cauchy sequence, hence converging since S is complete and Ω is to be intended with the induced metric.
By the arbitrariness of the sequence $(\tau_k)_k$ we deduce the limit $\lim_{\tau\to\bar{\tau}} \gamma^*(\tau)$ exists. □

In this section we have essentially proved that a complete space-like surface immersed in $\mathbb{M}^{2,1}$ is actually the graph of a function f satisfying $|\nabla f|^2 < 1$. A partial converse also holds: the graph of a locally Lipschitz function whose gradient has norm strictly smaller than 1 is a space-like surface, if it is complete.

The Gauss map in $\mathbb{M}^{2,1}$

Since the ambient metric g is nondegenerate and each tangent plane T_pS is space-like, the restriction of g to T_pS is positive-definite, hence nondegenerate. It follows that, for every $p \in S$,

$$\mathbb{M}^{2,1} = T_p\mathbb{M}^{2,1} = (T_pS)^\perp \oplus T_pS$$

Since S has codimension 1 in $\mathbb{M}^{2,1}$, the orthogonal complement $(T_pS)^\perp$ is one-dimensional. Therefore it is generated by a normal vector N_p to S.

We can easily show that N must be a time-like vector, since T_pS is space-like.
Let's fix $e_1, e_2 \in T_pM$ two independent vectors. Since N belongs to $(T_pM)^\perp$ we have $\langle e_1, N\rangle = \langle e_2, N\rangle = 0$. This implies that when we look at the representative matrix of g_p written in the basis (N, e_1, e_2), we have

$$[g_p] = \begin{pmatrix} \langle N, N\rangle & 0 & 0 \\ 0 & \langle e_1, e_1\rangle & \langle e_1, e_2\rangle \\ 0 & \langle e_2, e_1\rangle & \langle e_2, e_2\rangle. \end{pmatrix} =: \begin{pmatrix} \langle N, N\rangle & 0 \\ 0 & [(g|_S)_p] \end{pmatrix}.$$

Where the block $[(g|_S)_p]$ represents the form g restricted to the subspace T_pM. We conclude that $\langle N, N\rangle$ is negative just by observing that $\det[g_p] < 0$ as g is a Lorentzian metric, while $\det[(g|_S)_p] > 0$ since g becomes a metric when restricted to S.

We can now repeat the classical construction of the Gauss map in the Lorentzian setting. For every $p \in S$, the orthogonal complement $(T_pS)^\perp$ is a one-dimensional timelike subspace of $T_p\mathbb{M}^{2,1}$. Hence it contains exactly two unit vectors, characterized by the condition $\langle w, w\rangle = -1$. These two vectors lie on the two connected components of $\hat{\mathbb{H}}^2 = \{v \in \mathbb{M}^{2,1} : \langle v, v\rangle = -1\}$. By choosing the connected component

$$\mathbb{H} = \mathbb{H}^2 = \{v \in \mathbb{M}^{2,1} : \langle v, v\rangle = -1,\ v^3 > 0\},$$

we select, at each point $p \in S$, a unique unit normal vector. What is sometimes called a "time-orientation" for the space $\mathbb{M}^{2,1}$. This way we obtain a globally defined map

$$u : S \to \mathbb{H},$$

called the Gauss map of S.
This construction is analogous to the classical one, except that the target is $\mathbb{H}^2$ instead of $\mathbb{S}^2$. Moreover, unlike the Euclidean case, for a space-like surface we always have a globally defined unit normal field: once a time orientation of $\mathbb{M}^{2,1}$ is fixed, the timelike line $(T_pS)^\perp$ intersects the corresponding sheet of $\hat{\mathbb{H}}^2$ in exactly one point for every $p \in S$. Hence the Gauss map is globally well defined on the whole surface. Furthermore, the orientation of $\mathbb{M}^{2,1}$ induces an orientation on S.

$\mathbb{H}$ is a model for the Hyperbolic plane.
The space $\mathbb{H}$ that we just defined is clearly a surface immersed in $\mathbb{M}^{2,1}$, but there is more. First of all, we observe that, by manipulating the definition of $\mathbb{H}$ we can write it as the graph of the function $\varphi(x, y) = \sqrt{x^2 + y^2 + 1}$. The gradient of such function is $\nabla\varphi = \frac{1}{\sqrt{x^2+y^2+1}}(x, y)$, from which it's clear that $|\nabla\varphi| < 1$. By applying the result we proved before, we obtain that the bilinear form that $\mathbb{H}$ inherited from the immersion in $\mathbb{M}^{2,1}$ is a Riemannian metric.
In fact this space is one of the many models of the hyperbolic plane. Amongst these models there is one which might be more useful than others, the Poincaré disk. As a manifold, the Poincaré disk is simply the unit ball in $\mathbb{R}^2$, $\mathbb{D} = \{\xi \in \mathbb{R}^2 \mid |\xi| \leq 1\}$, while we can construct the metric by exhibiting a diffeomorphism between $\mathbb{D}$ and $\mathbb{H}$ pulling back the metric.
A possible diffeomorphism is the map p, whose inverse is the map q, as can be verified by direct computation:

$$\begin{aligned} p\colon \mathbb{H} &\longrightarrow \mathbb{D} \\ (x, y, t) &\longmapsto \frac{1}{1+t}(x, y) \end{aligned} \qquad\qquad \begin{aligned} q\colon \mathbb{D} &\longrightarrow \mathbb{H} \\ \xi &\longmapsto \frac{1}{1-|\xi|^2}(2\xi^1, 2\xi^2, 1 + |\xi|^2). \end{aligned}$$

It's clear that both functions are smooth, since $|\xi| < 1$ and $t > 1$.
We would like to find the expression of $q^*(g_{\mathbb{H}})$.To simplify the calculations we might observe that the metric of $\mathbb{H}$ inherited by $\mathbb{M}^{2,1}$ is invariant under a rotation on the t axis, and that such a rotation corresponds under p to a rotation of the disk. This means that the metric $q^*(g_H)$ will be invariant under rotation.
It is then sufficient to fix a certain $\xi \in \mathbb{D}$ of the form $\xi = (a, 0)$ and calculate $q^*(g_{\mathbb{H}})_\xi$; this will specify the whole metric tensor. By direct computation we obtain

$$[dq_\xi] = \frac{2}{(1-|\xi|^2)^2} \begin{pmatrix} 1+(\xi^1)^2-(\xi^2)^2 & 2\xi^1\xi^2 \\ 2\xi^1\xi^2 & 1-(\xi^1)^2+(\xi^2)^2 \\ 2\xi^1 & 2\xi^2 \end{pmatrix}.$$

Which, evaluated at $\xi = (a, 0)$, yields

$$[dq_{(a,0)}] = \frac{2}{1-a^2} \begin{pmatrix} \frac{1+a^2}{1-a^2} & 0 \\ 0 & 1 \\ \frac{2a}{1-a^2} & 0 \end{pmatrix} =: \frac{2}{1-a^2} \left(v_1 | v_2\right).$$

We observe that, with respect to the metric of $\mathbb{M}^{2,1}$, it holds

$$\langle v_1, v_1 \rangle = 1, \qquad \langle v_2, v_2 \rangle = 1, \qquad \langle v_1, v_2 \rangle = 0.$$

Which means that the representative matrix of $q^*(g_{\mathbb{H}})$ is a multiple of the identity, $[q^*(g_{\mathbb{H}})_{(0,a)}] = \frac{2}{1-a^2} I$. By the argument on rotational invariance, we conclude that for every $\xi \in \mathbb{D}$,

$$q^*(g_{\mathbb{H}})_\xi = \left(\frac{2}{1-|\xi|^2}\right)^2 g^E_\xi.$$

The disk $\mathbb{D}$ with this metric is called the Poincaré model of the Hyperbolic plane. A nice property of this model is to be what is called a "conformal" model of the hyperbolic plane, namely an immersed manifold in a Euclidean space whose metric is a multiple of the standard one.

What we discussed in this paragraph lets us identify $\mathbb{H}$ and $\mathbb{D}$. From now on we will use $\mathbb{D}$. For example, we will always use this space as the target of the Gauss map,

$$u : S \to \mathbb{D}.$$

C The Gauss-Codazzi equations and the curvatures of S

C.1 First and second fundamental forms

C.2 Mean curvature and Gaussian curvature expr

D Riemann surfaces and harmonic maps

We here introduce some notation and some fundamental concepts of Riemann surfaces. The most crucial part will be the fact that each metric on a surface is locally conformal to the Euclidean, the relationship between a Riemannian metric and a complex structure, the Uniformization theorem and some properties of Harmonic functions.

D.1 Preliminaries on Riemann surfaces

Complexification Let us fix M a smooth manifold and $\pi_E : E \to M$ a smooth vector bundle on M. We define the complexification of the tangent bundle TM as

$$T_{\mathbb{C}}M := TM \otimes \mathbb{C},$$

where the tensor product has to be intended fiberwise. TODO

Complex surfaces
Let us fix S a surface (a topological manifold of dimension 2).

Definition D.1. A complex atlas $\mathcal{C}$ is a collection of charts

$$\mathcal{C} = \{z_\alpha : U_\alpha \to \mathbb{C}\}$$

that covers the whole surface S and whose transition maps

$$z_\beta \circ z_\alpha^{-1} : \ z_\alpha\big(U_\alpha \cap U_\beta\big) \to z_\beta\big(U_\alpha \cap U_\beta\big)$$

are holomorphic as maps between open subsets of $\mathbb{C}$. We call $\mathscr{C}$ a complex or conformal structure if it is an atlas maximal under the subset relation. Finally we call the pair $(S, \mathscr{C})$ (just S for short) a Riemann surface.

Definition D.2. A continuous map $h : S_1 \to S_2$ between Riemann surfaces is said to be holomorphic if, in local coordinates $\{U_\alpha, z_\alpha\}$ on S_1 and $\{U'_\beta, z'_\beta\}$ on S_2,all the maps $z'_\beta \circ h \circ z_\alpha^{-1}$ are holomorphic wherever they are defined. We call f a biholomorphism if it is holomorphic, bijective and its inverse f^{-1} is holomoprhic.

Conformal metrics

Definition D.3. We say that two metrics g and g' on the surface S are conformal to each other if there exists a smooth function u such that

$$g' = e^u g.$$

Or in less formal terms, the metric g' is at each point multiple of g, by the means of a smooth function. The term e^u is called conformal factor.

Definition D.4. A map $f : (M, g) \to (N, h)$ between two Riemannian manifolds is conformal if it's a diffeomorphism and the pull back metric is conformal to the ambient metric, that is, there exists a smooth function u such that

$$f^*(h) = e^u g.$$

The following fact might be useful TODO

Lemma 1. *Given $\Omega, \Omega' \subset \mathbb{C}$ open subsets of $\mathbb{C}$, equipped with the flat metric, then a map $f : \Omega \to \Omega'$ is conformal iff it's a biholomorphism.*

Conformal structures on 2-surfaces
Here we state the results that lead to the fact that every Riemannian surface admits a complex structure.

Theorem D.1 (Gauss)**.** *Given g a Riemannian metric on S differentiable surface, and a point $p \in S$, there is a chart near p, $\varphi : U \to \mathbb{R}^2$ such that g takes the form and a smooth positive function λ such that*

$$g = \lambda^2(dx^2 + dy^2),$$

where (x, y) are the coordinates defined by φ.

Definition D.5. The chart (φ, U) in the theorem is called conformal, while the coordinates (x, y) on U are called isothermal coordinates around p.
:w

D.2 Uniformization theorem

D.3 Harmonic maps between Riemann surfaces

D.4 Energy density and Jacobian

D.5 Hopf differential

D.6 Holomorphicity of the Hopf differential

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E CMC surfaces and harmonic maps

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F Classification of hyperbolic CMC cuts

F.1 The mean curvature of the CMC immersed in $\mathbb{M}^{2,1}$ cannot be -1

F.2 The idea behind the classification

F.3 The map $\mathcal{A}$

F.3.1 Hyperbolic CMC cuts and conformal type of the disk

F.3.2 From a surface to a holomorphic quadratic differential

F.4 inverting the map $\mathcal{A}$

F.4.1 Reconstruction from the PDE data

F.4.2 Use of Gauss and Codazzi equations

F.4.3 Fundamental theorem of surface theory

F.5 Statement of the classification theorem

G Uniqueness for the PDE

So far we have reduced our classification problem to proving existence and uniqueness for the following differential problem.

Given q a smooth function on the Poincaré disk $\mathbb{D}$, find a smooth function

$$w : \mathbb{D} \to \mathbb{R}$$

such that

$$\Delta_p w = e^{2w} - q^2 e^{-2w} - 1 \ , \tag{1}$$ {eq:box}

where Δ_p is the Laplacian with respect to the metric ds_p^2 and such that

$$e^{2w} ds_p^2$$

is a complete metric on D, with

$$e^{2w} - q^2 e^{-2w} > 0 \ . \tag{2}$$ {eq:box2}

on $\mathbb{D}$.

rimuoverò questo box una volta scritte le sezioni precedenti

We will need the following generalized maximum principle, due to Cheng and Yau and found in the article [1].

G.1 A generalized maximum principle

Let M be a complete Riemannian manifold of dimension n, and fix a point $p \in M$. Assume moreover that M is simply connected and has nonpositive sectional curvature. This assumption is stronger than what is strictly needed in this section, but it greatly simplifies the discussion and is satisfied in the case of a complete space-like surface of $\mathbb{M}^{2,1}$. These hypotheses allow us to apply the Cartan-Hadamard theorem, so that the exponential map $\exp_p$ is a diffeomorphism. In particular, this tells us that the cut locus of p is empty, since for every $x \in M \setminus \{p\}$ the minimizing geodesic joining p to x is radial and therefore unique. It follows that the function $r(x) := d(x, p)$ is differentiable on $M \setminus \{p\}$ and that, for every $x \in M$, there exists a unique minimizing unit-speed geodesic

$$\sigma : [0, l] \to M$$

joining p to x, where $l = d(x, p)$. We denote by $\sigma'(t)$ its velocity vector, and by $\mathrm{Ric}(\sigma'(t))$ the Ricci curvature along σ. We then define

$$K(x) = \min_{0 \le k \le l} \left[\frac{n-1}{l-k} - \frac{1}{(l-k)^2} \int_k^l (t-k)^2 \, \mathrm{Ric}(\sigma'(t)) dt \right] \ . \tag{3}$$ {eq:intromaxprin1}

We now claim that, if $r(x) \ge 1$ and the Ricci curvature is bounded from below, then $K(x)$ is bounded from above by a constant independent of x. Indeed, if $\mathrm{Ric} \ge \mathscr{S}$, then

$$\frac{1}{(l-k)^2} \int_k^l (t-k)^2 \, \mathrm{Ric}(\sigma'(t)) \, dt \ge \frac{1}{(l-k)^2} \int_k^l (t-k)^2 \mathscr{S} \, dt = \frac{(l-k)}{3} \mathscr{S} \ .$$

Therefore in the argument of min in (3) we have

$$\frac{n-1}{l-k} - \frac{1}{(l-k)^2}\int_k^l (t-k)^2 \operatorname{Ric}(\sigma'(t))\,dt \le \frac{n-1}{l-k} - \frac{(l-k)}{3}\mathscr{S} \quad .$$

Since $r(x) = l \ge 1$, if we choose $k = l-1$ we get

$$K(x) \le n-1-\frac{\mathscr{S}}{3} \quad .$$

Which proves our claim.

The next lemma, due to Yau, provides the estimate for the distance function that will be used in the proof of Theorem G.1.

{lm:distest}

Lemma 2. *With the notation introduced above, for every $x \in M \setminus \{p\}$ such that $r(x) \ge 1$ we have*

$$\Delta r(x) \le K(x) \ .$$

Proof. We redirect the reader to [2] for a proof. □

We are ready to state and prove the maximum principle.

{th:maxprin}

Theorem G.1. *Let M be a complete, simply connected Riemannian manifold of nonpositive sectional curvature, with Ricci curvature bounded from below. Let $f \in C^2(M)$ be bounded from above. Fix $p \in M$, and let $r(x) = d(p,x)$. Then there exist a constant $C > 0$ and a sequence $(x_k)_k \subset M$ such that*

$$f(x_k) \to \sup_M f, \qquad |\nabla f|(x_k) \le \frac{C}{k}, \qquad \Delta f(x_k) \le \frac{C}{k}.$$

In particular,

$$\limsup_{k\to\infty} \Delta f(x_k) \le 0.$$

Proof. Let's define the auxiliary functions $u(x) = (f(x) - f(p) + 1)$ and $L(x) = \log(r^2(x)+2)$. Then let's fix a positive integer k and define the function g as

$$g(x) = \frac{f(x) - f(p) + 1}{(\log(r^2(x)+2)^{\frac{1}{k}}} = \frac{u(x)}{L(x)^{\frac{1}{k}}} \quad .$$

Let's observe that $g(p) = \frac{1}{\log(2)^{\frac{1}{k}}} > 0$ and that $\lim_{r(x)\to\infty} g(x) = 0$. Since g is continuous, this means that it attains its maximum at a certain point x_k.

Let's show first that

$$\limsup_{k\to\infty} f(x_k) = \sup_M f \quad . \tag{4}$$

{eq:boundprin13}

Assume it by contradiction to be false. Then by definition of sup and lim sup there exists a $\delta > 0$ and a $\bar{x} \in M$ such that for k sufficiently big,

$$f(x_k) < f(\bar{x}) - \delta \ , \qquad \text{equivalently,} \qquad f(x_k) + \delta < f(\bar{x}) \quad . \tag{5}$$

{eq:boundprin12}

We have two possibilities, either $r(x_k) \overset{k\to\infty}{\longrightarrow} \infty$ or up to taking a subsequence it is bounded.
If $r(x_k) \overset{k\to\infty}{\longrightarrow} \infty$, then we have, for k sufficiently large, $r(x_k) > r(\bar{x})$ and $f(\bar{x}) > f(x_k)$ so that

$$g(\bar{x}) = \frac{f(\bar{x}) - f(p) + 1}{(\log(r^2(\bar{x}) + 2)^{\frac{1}{k}}} > \frac{f(x_k) - f(p) + 1}{(\log(r^2(x_k) + 2)^{\frac{1}{k}}} = g(x_k) \quad .$$

Which contradicts the maximality of x_k.

If on the other hand $r(x_k)$ is a bounded sequence, we can find a closed geodesic ball $\overline{B}(p;\varrho)$ with $\varrho > 0$ that contains x_k for every k sufficiently big. Since M is complete, the Hopf Rinow theorem implies that $\overline{B}(p;\varrho)$ is compact. Therefore, after passing to a subsequence, we may assume that

$$x_k \to x_0 \quad .$$

By taking the limit in (5) we have

$$f(x_0) + \delta \leq f(\bar{x}) \quad .$$

Now we observe that

$$\frac{f(\bar{x}) - f(p) + 1}{(\log(r^2(\bar{x}) + 2)^{\frac{1}{k}}} \overset{k\to\infty}{\longrightarrow} f(\bar{x}) - f(p) + 1$$

and

$$\frac{f(x_k) - f(p) + 1}{(\log(r^2(x_k) + 2)^{\frac{1}{k}}} \overset{k\to\infty}{\longrightarrow} f(x_0) - f(p) + 1 \quad .$$

Since we know $f(\bar{x}) - f(p) + 1 > f(x_0) - f(p) + 1$, we obtain, for k sufficiently large,

$$\frac{f(\bar{x}) - f(p) + 1}{(\log(r^2(\bar{x}) + 2)^{\frac{1}{k}}} > \frac{f(x_k) - f(p) + 1}{(\log(r^2(x_k) + 2)^{\frac{1}{k}}} \quad .$$

Which again is a contradiction, hence proving (4).
It is obvious that, up to taking a subsequence, we can assume $f(x_k) \to \sup_M f$.

Let's now focus on the two estimates. As we already mentioned, the Cartan-Hadamard M theorem grants that x_k is not in the cut-locus of p, and so that the function $d(p,x)^2 = r(x)^2$ is differentiable in x_k.
Let's calculate the gradient of g,

$$\nabla g(x) = \frac{\nabla f(x)}{\log(r(x)^2 + 2)^{1/k}} - \frac{2(f(x) - f(p) + 1)r(x)}{k(r(x)^2 + 2)\log(r(x)^2 + 2)^{1+1/k}} \nabla r(x) \quad .$$

Since x_k is a maximum point of g_k, we have $\nabla g(x_k) = 0$. Hence

$$\nabla f(x_k) = \frac{2(f(x_k) - f(p) + 1)\, r(x_k)}{k\,(r^2(x_k) + 2)\,\log(r^2(x_k) + 2)} \nabla r(x_k) \ , \tag{6}$$ {eq:boundprin9}

$$|\nabla f|(x_k) = \frac{2(f(x_k) - f(p) + 1)\, r(x_k)}{k\,(r^2(x_k) + 2)\,\log(r^2(x_k) + 2)} \ , \tag{7}$$ {eq:boundprin8}

where we used $|\nabla r(x_k)| = 1$.
We then observe, since f is bounded from above, also u is

$$u(x) = (f(x) - f(p) + 1) \leq \sup_M f - f(p) + 1 =: A.$$

Also, it's not difficult to show that the real function satisfies

$$\frac{2x}{(x^2+2)\log(x^2+2)} \leq 2 \ , \quad . \tag{8}$$ {eq:boundprin14}

so that by substituting in (7) we get the desired estimate

$$|\nabla f|(x_k) \leq \frac{2A}{k} \quad .$$

Let's now focus on the estimate on the Laplacian. Again by direct computation we can find the expression of the Laplacian of g,

$$\begin{aligned}\Delta g = \frac{\Delta f}{L^{1/k}} - \frac{4r}{k(r^2+2)L^{1+1/k}}\langle \nabla f, \nabla r\rangle - \frac{2u}{k(r^2+2)L^{1+1/k}} \\ - \frac{2ur\,\Delta r}{k(r^2+2)L^{1+1/k}} + \frac{4ur^2}{k(r^2+2)^2L^{1+1/k}} + \frac{4(k+1)ur^2}{k^2(r^2+2)^2L^{2+1/k}}.\end{aligned}$$

Since x_k is a maximum point of g, we have $\Delta g(x_k) \leq 0$. Using the expression for Δg, the identity $\nabla g(x_k) = 0$, and

$$\langle \nabla f, \nabla r\rangle(x_k) = \frac{2u(x_k)r(x_k)}{k(r^2(x_k)+2)\log(r^2(x_k)+2)} \ ,$$

which we have derived from (6), we obtain

$$\begin{aligned}\Delta f(x_k) \leq \frac{2u(x_k)\big(1 + r(x_k)\Delta r(x_k)\big)}{k(r^2(x_k)+2)\log(r^2(x_k)+2)} - \frac{4u(x_k)r^2(x_k)}{k(r^2(x_k)+2)^2\log(r^2(x_k)+2)} \\ - \frac{4(k-1)u(x_k)r^2(x_k)}{k^2(r^2(x_k)+2)^2\log^2(r^2(x_k)+2)} \ .\end{aligned}$$

In particular, by discarding the last two nonpositive terms, we get

$$\Delta f(x_k) \leq \frac{2u(x_k)\big(1 + r(x_k)\Delta r(x_k)\big)}{k(r^2(x_k)+2)\log(r^2(x_k)+2)} \ . \tag{9}$$ {eq:boundprin10}

Now, we can reduce to one of the following two cases, by discarding the indexes which do not satisfy the corresponding condition:

$$r(x_k) < 1 \qquad \text{for every } k,$$

or

$$r(x_k) \geq 1 \qquad \text{for every } k \quad .$$

for some $x_0 \in \overline{B}(p;1)$.

Let's assume first that $r(x_k) < 1$ for every k. Then by definition $x_k \in \overline{B}(p;1)$ for every k. Again by the compactness of B we may assume, up to taking a subsequence, that

$$x_k \to x_0$$

for some $x_0 \in \overline{B}(p;1)$. But we know that $f(x_k) \to \sup f$ and that f is continuous, which implies that x_0 is a maximum point.
This entails that

$$\nabla f(x_0) = 0 \qquad \text{and} \qquad \Delta f(x_0) \leq 0 \ .$$

Which verifies trivially the thesis by setting $x_k \equiv x_0$.

Assume now that $r(x_k) \geq 1$ for every k. First, by observing that $1/[(x^2+2)\log(x^2+2)] \leq 2$, it follows from (9),

$$\begin{aligned}
\Delta f(x_k) &\leq \frac{2u(x_k)\big(1 + r(x_k)\Delta r(x_k)\big)}{k(r^2(x_k)+2)\log(r^2(x_k)+2)} \\
[u \leq A] \qquad &\leq \frac{2A}{k}\left(\frac{1}{(r^2(x_k)+2)\log(r^2(x_k)+2)} + \frac{r(x_k)\Delta r(x_k)}{(r^2(x_k)+2)\log(r^2(x_k)+2)}\right) \\
[(8)] \qquad &\leq \frac{2A}{k}\left(1 + 2\Delta r(x_k)\right) \ . \qquad (10)
\end{aligned}$$

{eq:boundprin11}

Then we apply the estimate of Lemma 2, which is valid since $r(x_k) \geq 1$,

$$\Delta r(x) \leq K(x) \ .$$

But, as we already observed, $K(x)$ is uniformly bounded from above when $r(x) \geq 1$ since the Ricci curvature is bounded from below.
So we conclude in (10) that there is a constant $\tilde{C}$ for which

$$\Delta f(x_k) \leq \frac{\tilde{C}}{k} \ .$$

□

A boundedness result
We will also use this theorem to derive a boundedness result for a certain differential inequality. We first make a few preliminary observations, and then state the theorem at the end of the section.

Let M be a Riemannian manifold with Ricci curvature bounded from below, and let $u \in C^2(M)$ satisfy

$$\Delta u \geq f \ , \qquad (11)$$

{eq:boundprin2}

where the assumptions on f will be specified later.
Now let $F \in C^2(\mathbb{R})$ be a positive function, whose precise form will also be given later. Since $F > 0$, the function $-1/F$ is bounded from above. We may therefore apply Theorem G.1 to $-1/F(u)$ and obtain a sequence $(x_k)_k$ such that

$$\lim_{k\to\infty} \frac{1}{F(u(x_k))} = \inf \frac{1}{F(u)} = -\sup\left(-\frac{1}{F(u)}\right). \qquad (12)$$

{eq:boundprin7}

Moreover, the corresponding gradient and Laplacian estimates hold:

$$\frac{|F'(u(x_k))|\,|\nabla u|(x_k)}{F(u(x_k))^2} \leq \frac{1}{k}, \qquad (13)$$

{eq:boundprin3}

$$-\frac{F''(u(x_k))\,|\nabla u|^2(x_k)}{F(u(x_k))^2} - \frac{F'(u(x_k))\,\Delta u(x_k)}{F(u(x_k))^2} + \frac{2(F'(u(x_k)))^2\,|\nabla u|^2(x_k)}{F(u(x_k))^3} \geq -\frac{1}{k}\ . \tag{14}$$ {eq:boundprin16}

For simplicity, from now on we evaluate all quantities at x_k.
Then, multiplying (14) by $\dfrac{F'(u)^2}{F(u)^2|F''(u)|}$, we obtain

$$-\frac{F''(u)}{|F''(u)|}\frac{F'(u)^2|\nabla u|^2}{F(u)^4} - \frac{F'(u)^3}{F(u)^4}\frac{\Delta u}{|F''(u)|} + \frac{2F'(u)^2}{F(u)|F''(u)|}\frac{F'(u)^2|\nabla u|^2}{F(u)^4} \geq -\frac{F'(u)^2}{kF(u)^2|F''(u)|}\ . \tag{15}$$ {eq:boundprin1}

We now estimate the terms on the left-hand side of (15). First,

$$-\frac{F'(u)^3}{F(u)^4}\frac{\Delta u}{|F''(u)|} \leq -\frac{F'(u)^3}{F(u)^4}\frac{f(u)}{|F''(u)|}\ . \qquad \text{[by (11)]}$$

Moreover,

$$-\frac{F''(u)}{|F''(u)|}\frac{F'(u)^2|\nabla u|^2}{F(u)^4} \leq \left|\frac{F'(u)^2|\nabla u|^2}{F(u)^4}\right| \leq \frac{1}{k^2}\ . \qquad \text{[by (13)]}$$

Also,

$$\begin{aligned}\frac{2F'(u)^2}{F(u)|F''(u)|}\frac{F'(u)^2|\nabla u|^2}{F(u)^4} &\leq 2\left|\frac{F'(u)^2}{F(u)|F''(u)|}\right|\left|\frac{F'(u)^2|\nabla u|^2}{F(u)^4}\right| \\ &\leq \frac{2}{k^2}\left|\frac{F'(u)^2}{F(u)|F''(u)|}\right|\ . \qquad \text{[by (13)]}\end{aligned}$$

Therefore, from (15) we obtain

$$\frac{1}{k^2} - \frac{F'(u)^3 f(u)}{F(u)^4|F''(u)|} + \frac{2}{k^2}\left|\frac{F'(u)^2}{F(u)|F''(u)|}\right| \geq -\frac{1}{k}\frac{F'(u)^2}{F(u)^2|F''(u)|}\ . \tag{16}$$ {eq:boundprin6}

Suppose now that F satisfies

$$\limsup_{t\to\infty}\frac{|F'(t)|^2}{F(t)\,|F''(t)|} < \infty\ , \tag{17}$$ {eq:boundprin4}

and

$$\liminf_{t\to\infty}\frac{F'(t)^3 f(t)}{F(t)^4\,|F''(t)|} > 0\ . \tag{18}$$ {eq:boundprin5}

We claim that this implies

$$\limsup_{k\to\infty} u(x_k) < \infty\ .$$

Indeed, suppose by contradiction that $\limsup_{k\to\infty} u(x_k) = \infty$. Passing to a subsequence if necessary we may assume that

$$u(x_k) \to \infty\ .$$

Taking $\limsup$ in (16), and using (17) together with the fact that later our choice of F will satisfy $F \geq 1$, we obtain

$$-\liminf_{k\to\infty} \frac{F'(u(x_k))^3 f(u(x_k))}{F(u(x_k))^4 |F''(u(x_k))|} \geq 0 \quad ,$$

that is,

$$\liminf_{k\to\infty} \frac{F'(u(x_k))^3 f(u(x_k))}{F(u(x_k))^4 |F''(u(x_k))|} \leq 0 \quad ,$$

which contradicts (18).

So we have proved so far that

$$\limsup_{k\to\infty} u(x_k) < \infty \quad .$$

Passing to a subsequence if necessary, we may therefore assume that

$$u(x_k) \to \ell$$

for some $\ell \in \mathbb{R}$.

Let us now show that, if we assume in addition that F is strictly increasing, then u is bounded from above.
Since $F > 0$, from (12) we obtain

$$\lim_{k\to\infty} \frac{1}{F(u(x_k))} = \inf \frac{1}{F(u)} = \frac{1}{\sup_M F(u)} \quad ,$$

and therefore

$$F(u(x_k)) \to \sup F(u) \quad .$$

On the other hand, by continuity of F and the fact that $u(x_k) \to \ell$, we also have

$$F(u(x_k)) \to F(\ell) \quad .$$

Hence

$$F(\ell) = \sup F(u) \quad .$$

Since F is strictly increasing, for every $x \in M$ we have

$$F(u(x)) \leq F(\ell) \qquad \Longrightarrow \qquad u(x) \leq \ell \quad .$$

This means that ℓ is an upper bound for u, and hence u is bounded from above:

$$\sup u < \infty \quad .$$

We have therefore reduced the problem to finding a function F which is positive, strictly increasing, and satisfies (17) and (18). Let us search for such a function of the form

$$F(s) = \int_a^s \left(\int_a^t g(\tau)\, d\tau \right)^{-\frac{1}{2}} dt + 1 \quad , \tag{19}$$ {eq:boundprin15}

where g is a positive continuous function defined on $[a, \infty)$ and such that

$$\int_b^\infty \left(\int_a^t g(\tau)\, d\tau \right)^{-\frac{1}{2}} dt < \infty$$

for some $b \geq a$. If we want F to be positive, the expression in (19) may fail to achieve that on the whole line. So we define it on $[a, \infty)$ and then extend F to all of $\mathbb{R}$ in such a way that it becomes a strictly increasing C^2 function.

Let's now translate conditions (17) and (18) into conditions on the function g. To do so, we first compute the relevant quantities.

$$F'(s) = \left(\int_a^s g(\tau)\, d\tau \right)^{-\frac{1}{2}} \quad ,$$

and therefore

$$F''(s) = -\frac{1}{2} \left(\int_a^s g(\tau)\, d\tau \right)^{-\frac{3}{2}} g(s) \quad .$$

Hence

$$\frac{|F'(t)|^2}{F(t)\,|F''(t)|} = 2\, \frac{\left(\int_a^t g(\tau)\, d\tau \right)^{1/2}}{F(t)\, g(t)} \quad ,$$

so condition (17) is equivalent to

$$\limsup_{t \to \infty} \frac{\left(\int_a^t g(\tau)\, d\tau \right)^{1/2}}{F(t)\, g(t)} < \infty \quad .$$

Moreover,

$$\frac{F'(t)^3 f(t)}{F(t)^4 |F''(t)|} = 2\, \frac{f(t)}{F(t)^4 g(t)} \quad ,$$

so condition (18) is equivalent to

$$\liminf_{t \to \infty} \frac{f(t)}{F(t)^4 g(t)} > 0 \quad .$$

Finally, we observe that since g is positive, if $s \leq t$, then

$$\left(\int_a^s g(\tau)\, d\tau \right)^{-\frac{1}{2}} \geq \left(\int_a^t g(\tau)\, d\tau \right)^{-\frac{1}{2}} \quad .$$

Therefore, for $t \geq b$,

$$\begin{aligned} F(t) - 1 &= \int_a^t \left(\int_a^s g(\tau)\, d\tau \right)^{-\frac{1}{2}} ds \\ &\geq \int_b^t \left(\int_a^s g(\tau)\, d\tau \right)^{-\frac{1}{2}} ds \\ &\geq \int_b^t \left(\int_a^t g(\tau)\, d\tau \right)^{-\frac{1}{2}} ds \\ &= (t-b) \left(\int_a^t g(\tau)\, d\tau \right)^{-\frac{1}{2}} \quad . \end{aligned}$$

Hence

$$F(t) \geq (t-b) \left(\int_a^t g(\tau)\, d\tau \right)^{-\frac{1}{2}} \quad .$$

This allows us to further estimate the quantity appearing in (17):

$$\begin{aligned} \frac{\left(\int_a^t g(\tau)\, d\tau \right)^{1/2}}{F(t)\, g(t)} &\leq \frac{\left(\int_a^t g(\tau)\, d\tau \right)^{1/2}}{(t-b) \left(\int_a^t g(\tau)\, d\tau \right)^{-1/2} g(t)} \\ &= \frac{\int_a^t g(\tau)\, d\tau}{(t-b) g(t)} \\ &\sim \frac{\int_a^t g(\tau)\, d\tau}{t\, g(t)} \quad . \end{aligned}$$

We have thus proved the following theorem.

{th:boundprin}

Theorem G.2. *Let M be a Riemannian manifold whose Ricci curvature is bounded from below, and let $u \in C^2(M)$ satisfy*

$$\Delta u \geq f(u) \quad .$$

A sufficient condition for u to be bounded from above on M is the existence of a continuous positive function g, defined on an interval $[a, \infty)$, such that

- $\displaystyle\int_b^\infty \left(\int_a^t g(\tau)\, d\tau \right)^{-1/2} dt < \infty$ *for some* $b \geq a$,
- $\displaystyle\limsup_{t \to \infty} \frac{\int_a^t g(\tau)\, d\tau}{t\, g(t)} < \infty$,
- $\displaystyle\liminf_{t \to \infty} \frac{f(t)}{g(t)} > 0$.

G.2 The uniqueness theorem

In order to prove uniqueness, we pass to an auxiliary formulation of the problem. We first prove uniqueness in this new setting, and then show that this implies

uniqueness for the original equation.
Recall that we are looking for a function w such that the metric g can be written as

$$g = e^{2w} ds_p^2,$$

where ds_p^2 is the Poincaré metric on $\mathbb{D}$. Since ds_p^2 is itself conformal to the Euclidean metric $|dz|^2$, with conformal factor

$$\frac{4}{(1-|z|^2)^2},$$

it follows that g is also conformal to $|dz|^2$. In fact, we can write

$$g = \frac{4e^{2w}}{(1-|z|^2)^2}|dz|^2.$$

If we now express the conformal factor in the form $e^{2\tilde{w}}$, we find a transformation between w and $\tilde{w}$,

$$\tilde{w} = w + \log\left(\frac{2}{1-|z|^2}\right).$$

This lets us write the metric g as

$$g = e^{2\tilde{w}}|dz|^2.$$

Therefore w and $\tilde{w}$ describe the same metric, the former with respect to the Poincaré background metric and the latter with respect to the Euclidean one. We perform this transformation as it will be convenient to study the corresponding equation in the Euclidean setting first, where the relevant operator is the standard Laplacian Δ. Once we have proved uniqueness there, we will translate the result back to the original formulation.

Theorem G.3. *Let h be a continuous function defined on a domain $\Omega \subset \mathbb{C}$. Let $\tilde{w}$ be a solution of*

$$\Delta\tilde{w} = e^{2\tilde{w}} - h^2 e^{-2\tilde{w}} \tag{20}$$ {eq:unithm}

in Ω, and assume that

$$\Delta\tilde{w} \geq 0 \quad ,$$

and that $e^{2\tilde{w}}|dz|^2$ is a complete metric on Ω. Then any other solution v of the equation satisfies $v \leq \tilde{w}$.

Proof. Let us define $\eta = v - \tilde{w}$. We need to prove that $\eta \leq 0$.
We have

$$\begin{aligned}\Delta\eta &= \Delta v - \Delta\tilde{w} \\ &\overset{[(20)]}{=} e^{2v} - h^2 e^{-2v} - e^{2\tilde{w}} + h^2 e^{-2\tilde{w}} \quad .\end{aligned} \tag{21}$$ {eq:unithm2}

Let's now try to explicit $\tilde{w}$ and η in the RHS of (21),

$$e^{2v} = e^{2(\eta+\tilde{w})} = e^{2\eta}e^{2\tilde{w}} \quad .$$

Let's define the quantity $|\mu|^2 = h^2 e^{-4\tilde{w}}$, we have then

$$h^2 e^{-2v} = h^2 e^{-2(\eta+\tilde{w})} = h^2 e^{-2\tilde{w}} e^{-2\eta} = |\mu|^2 e^{2\tilde{w}} e^{-2\eta},$$

and

$$h^2 e^{-2\tilde{w}} = \left(h^2 e^{-4\tilde{w}}\right) e^{2\tilde{w}} = |\mu|^2 e^{2\tilde{w}} \quad .$$

So, by substituting what we just found in (21), we get

$$\begin{aligned} \Delta\eta &= e^{2\tilde{w}} e^{2\eta} - |\mu|^2 e^{2\tilde{w}} e^{-2\eta} - e^{2\tilde{w}} + |\mu|^2 e^{2\tilde{w}} \\ &= e^{2\tilde{w}} (e^{2\eta} - |\mu|^2 e^{-2\eta} - 1 + |\mu|^2) \quad . \end{aligned} \tag{22}$$ {eq:unithm3}

The Laplacian with respect to the metric $g = e^{2\tilde{w}}|dz|^2$ is simply $e^{-2\tilde{w}}\Delta$. So in (22) we obtain

$$\Delta_g \eta = e^{-2\tilde{w}} \Delta\eta = e^{2\eta} - |\mu|^2 e^{-2\eta} - 1 + |\mu|^2 \quad . \tag{23}$$ {eq:unithm5}

Let us now observe that $|\mu|^2 \leq 1$. Indeed, by (20),

$$e^{2\tilde{w}} - h^2 e^{-2\tilde{w}} \geq 0 \quad ,$$

hence

$$e^{2\tilde{w}} (1 - h^2 e^{-4\tilde{w}}) \geq 0 \quad ,$$

that is,

$$1 - |\mu|^2 \geq 0 \quad ,$$

so $|\mu|^2 \leq 1$.
Therefore, from (23) we get

$$\begin{aligned} \Delta_g \eta &= e^{2\eta} - |\mu|^2 e^{-2\eta} - 1 + |\mu|^2 \\ &\geq e^{2\eta} - e^{-2\eta} - 1 =: f(\eta) \quad . \end{aligned}$$

We now apply Theorem G.2 to the function η on the complete simply connected Riemannian surface (Ω, g). For the auxiliary function appearing in the theorem, we may take

$$\gamma(t) = t^2$$

on $[1, \infty)$. A direct computation shows that γ satisfies the required assumptions, since

$$\int_b^\infty \left(\int_1^t \tau^2 \, d\tau \right)^{-1/2} dt < \infty, \qquad \limsup_{t\to\infty} \frac{\int_1^t \tau^2 \, d\tau}{t^3} < \infty,$$

and

$$\liminf_{t\to\infty} \frac{f(t)}{t^2} = \liminf_{t\to\infty} \frac{e^{2t} - e^{-2t} - 1}{t^2} > 0 \quad .$$

It remains only to check that the Ricci curvature of g is bounded from below which, since we are on a surface, is the same as the Gaussian curvature. For the metric

$$g = e^{2\tilde{w}} |dz|^2 \quad ,$$

the Gaussian curvature is

$$\begin{aligned} K(g) &= -e^{-2\tilde{w}}\Delta\tilde{w} \\ &= -e^{-2\tilde{w}}\left(e^{2\tilde{w}} - h^2 e^{-2\tilde{w}}\right) \\ &= -1 + h^2 e^{-4\tilde{w}} \\ &= -1 + |\mu|^2 \quad . \end{aligned}$$

Since $|\mu|^2 \leq 1$, it follows that

$$-1 \leq K(g) \leq 0 \quad ,$$

and in particular the curvature is bounded from below. Therefore we can apply Theorem G.2 and conclude that η is bounded from above.
Now we can also apply Theorem G.1 to η, obtaining a sequence $(z_k)_k \subset \Omega$ such that

$$\eta(z_k) \to \bar{\eta} = \sup_{\Omega} \eta, \qquad \limsup_{k\to\infty} \Delta_g \eta(z_k) \leq 0 \quad .$$

Since $|\mu|^2$ is continuous and bounded, by passing to a subsequence if necessary we may also assume that

$$|\mu|^2(z_k) \to a$$

for some $a \leq 1$. Taking limits in (23), we obtain

$$\begin{aligned} 0 &\geq \limsup_{k\to\infty} \Delta_g \eta(z_k) \\ &= \lim_{k\to\infty} \left(e^{2\eta(z_k)} - |\mu|^2(z_k) e^{-2\eta(z_k)} - 1 + |\mu|^2(z_k)\right) \\ &= e^{2\bar{\eta}} - a e^{-2\bar{\eta}} - 1 + a \\ &=: \varphi(\bar{\eta}) \quad . \end{aligned} \tag{24}$$

{eq:unithm4}

We have defined in the last expression the function $\varphi(s) = e^{2s} - ae^{-2s} - 1 + a$, which has strictly positive derivative $\varphi'(s) = 2e^{2s} + 2ae^{-2s} > 0$, so that it's strictly increasing. Moreover $\varphi(0) = 0$, hence (24) implies

$$\bar{\eta} \leq 0, \qquad \text{therefore, on } \Omega, \qquad \eta \leq 0 \quad .$$

This finally implies $v \leq \tilde{w}$. □

{cor:uni1}

Corollary G.1. *Let h be a continuous function defined on a domain $\Omega \subset \mathbb{C}$ and let $\tilde{w}_1, \tilde{w}_2$ two solutions of*

$$\Delta\tilde{w} = e^{2\tilde{w}} - h^2 e^{-2\tilde{w}} \geq 0 \tag{25}$$

in Ω such that $e^{2\tilde{w}_i}$ are complete metrics on Ω. Then $\tilde{w}_1 \equiv \tilde{w}_2$.

Proof. Just observe that for the previous theorem, $\tilde{w}_1 \leq \tilde{w}_2$ and $\tilde{w}_2 \leq \tilde{w}_1$. □

{cor:uni2}

Corollary G.2. *There is at most one solution of the equation* (1) *such that the associated metric is complete.*

Proof. Let w_1, w_2 be two solutions of

$$\Delta_p w = e^{2w} - q^2 e^{-2w} - 1$$

such that (1) holds and the metrics $e^{2w_1} ds_p^2$ and $e^{2w_2} ds_p^2$ are complete.

As we mentioned in the introduction, we can translate this problem into one involving the Euclidean Laplacian by means of the following transformation: for $i = 1, 2$, define

$$\tilde{w}_i = w_i + \log\left(\frac{2}{1-|z|^2}\right) \quad . \tag{26}$$ {eq:unithm6}

Since

$$ds_p^2 = \frac{4}{(1-|z|^2)^2}|dz|^2,$$

we have

$$e^{2\tilde{w}_i}|dz|^2 = e^{2w_i}\frac{4}{(1-|z|^2)^2}|dz|^2 = e^{2w_i} ds_p^2.$$

Hence $e^{2\tilde{w}_i}|dz|^2$ is complete.

Let us now show that each $\tilde{w}_i$ solves equation (1) and satisfies (2). First we recall that

$$\Delta_p = \frac{(1-|z|^2)^2}{4}\Delta$$

and observe that

$$\Delta \log\left(\frac{2}{1-|z|^2}\right) = \frac{4}{(1-|z|^2)^2} \quad .$$

So it follows that

$$\begin{aligned}
\Delta\tilde{w}_i &= \Delta w_i + \Delta \log\left(\frac{2}{1-|z|^2}\right) \\
&= \frac{4}{(1-|z|^2)^2}\Delta_p w_i + \frac{4}{(1-|z|^2)^2} \\
&= \frac{4}{(1-|z|^2)^2}\left(e^{2w_i} - q^2 e^{-2w_i} - 1\right) + \frac{4}{(1-|z|^2)^2} \\
&= \frac{4}{(1-|z|^2)^2}e^{2w_i} - \frac{4}{(1-|z|^2)^2}q^2 e^{-2w_i}.
\end{aligned} \tag{27}$$ {eq:unithm7}

Now, from (26) we get

$$\frac{4}{(1-|z|^2)^2}e^{2w_i} = e^{2\tilde{w}_i}, \qquad e^{-2w_i} = \frac{4}{(1-|z|^2)^2}e^{-2\tilde{w}_i}.$$

Therefore, substituting in (27), we obtain

$$\Delta\tilde{w}_i = e^{2\tilde{w}_i} - \left(\frac{4q}{(1-|z|^2)^2}\right)^2 e^{-2\tilde{w}_i} \quad . \tag{28}$$ {eq:unithm8}

Define

$$h(z) = \frac{4}{(1-|z|^2)^2}q(z).$$

Then h is continuous on $\mathbb{D}$, and (28) becomes

$$\Delta \tilde{w}_i = e^{2\tilde{w}_i} - h^2 e^{-2\tilde{w}_i} \quad . \tag{29}$$ {eq:unithm9}

Moreover, since (2) holds for w_i, we have

$$e^{2w_i} - q^2 e^{-2w_i} > 0,$$

and

$$\Delta \tilde{w}_i > 0.$$

This means that $\tilde{w}_1$ and $\tilde{w}_2$ are two solutions of (1) and (2) with complete metrics $e^{2\tilde{w}_i}|dz|^2$. Finally we apply corollary G.1 to conclude that $\tilde{w}_1 = \tilde{w}_2$. Hence, by (26), we conclude that

$$w_1 = w_2.$$

□

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H existence for the nonlinear PDE